Problem 49 (JBMO 2019 Shortlist N6). Find all triples (a, b, c) of nonnegative integers that satisfy

$$a! + 5^b = 7^c.$$

Solution. The solutions are $(a, b, c) = (2, 1, 1), (3, 0, 1), (4, 2, 2)$.

Since 5^b and 7^c are odd, $a!$ must be even and hence $a \geqslant 2$.

When $a = 2, 3$, if $b \geqslant 2$, then $7^c \equiv 2, 6 \pmod{25}$. This is impossible since $7^c \equiv 7, 24, 18, 1 \pmod{25}$. By checking $b = 0, 1$, we find the only solutions $(a, b, c) = (2, 1, 1), (3, 0, 1)$.

When $a = 4$, we have $5^b \equiv 7^c \pmod 8$. Since $5^b \equiv 5, 1 \pmod 8$ and $7^c \equiv 7, 1 \pmod 8$, we must have $5^b \equiv 7^c \equiv 1 \pmod 8$. So both b and c are even. Let $b = 2m$ and $c = 2n$. Then

$$24 = 7^c - 5^b = (7^n)^2 - (5^m)^2 = (7^n - 5^m)(7^n + 5^m).$$

Note that $7^n - 5^m$ and $7^n + 5^m$ have the same parity. So $(7^n - 5^m, 7^n + 5^m) = (2, 12), (4, 6)$. One checks that the only solution is $(m, n) = (1, 1)$, i.e. $(b, c) = (2, 2)$.

When $a \geqslant 5$, we have $5^b \equiv 7^c \pmod 5$. Since $7^c \not\equiv 0 \pmod 5$, we need $b = 0$. Now, if $a \geqslant 7$, then $7^c \equiv 1 \pmod 7$, which forces $c = 0$. There is no solution. By checking $a = 5, 6$, we find no solution.

Therefore, the solutions are $(a, b, c) = (2, 1, 1), (3, 0, 1), (4, 2, 2)$. □

Problem 50 (Brazil MO 2014 Problem 2). Find all integers $n > 1$ such that for each $r = 0, 1, 2, \ldots, n-1$, there exists a multiple of n whose sum of digits leaves a remainder r when divided by n.

Solution. The answer is any positive integer $n > 1$ not divisible by 3.

If $3 \mid n$, then any multiple of n has sum of digits divisible by 3, hence there is no such integer with $r = 1$.

If $3 \nmid n$, then $(n, 9) = 1$ and we can find a positive integer a such that $9a \equiv -r \pmod n$. Also, we can find a positive integer b such that $a + b \equiv r \pmod n$. Let $n = 2^x 5^y m$ where $(m, 10) = 1$. Consider the number

$$k = 10^{x+y}(10^{\varphi(m)} + 10^{2\varphi(m)} + \cdots + 10^{b\varphi(m)} + 10^{\varphi(m)+1} + 10^{2\varphi(m)+1} + \cdots + 10^{a\varphi(m)+1}). \tag{1}$$

By Euler-Fermat theorem, we have

$$k \equiv 10^{x+y}(1 + 1 + \cdots + 1 + 10 + 10 + \cdots + 10) = 10^{x+y}(b + 10a) \equiv 10^{x+y}(-r + r) = 0 \pmod m.$$

Also, $2^x 5^y \mid k$. Therefore, $n \mid k$. On the other hand, each term in (1) corresponds to a distinct digit. Therefore, the sum of digits of k is $a + b$, which is congruent to r modulo n. This shows k satisfies the requirement. □

Problem 51 (Canada MO 2010 Problem 5). Let $P(x)$ and $Q(x)$ be polynomials with integer coefficients. Let $a_n = n! + n$. Show that if $\dfrac{P(a_n)}{Q(a_n)}$ is an integer for every n, then $\dfrac{P(n)}{Q(n)}$ is an integer for every integer n such that $Q(n) \neq 0$.

Solution. Let $P(x) = Q(x)A(x) + R(x)$ for some $A(x), R(x) \in \mathbb{Q}[x]$ where $\deg R < \deg Q$. Let d be the least common multiple of the denominators of all coefficients of $A(x)$. Then $B(x) = dA(x) \in \mathbb{Z}[x]$. As $dP(a_n) = Q(a_n)B(a_n) + dR(a_n)$ where $P(a_n), Q(a_n) \in \mathbb{Z}$, we have $dR(a_n) \in \mathbb{Z}$. Also, as $Q(a_n) \mid P(a_n)$, this implies $Q(a_n) \mid dR(a_n)$. Note that a_n can be arbitrarily large, and $|Q(x)| > |dR(x)|$ for sufficiently large x since $\deg Q > \deg R$. Therefore, we must have $R(a_n) = 0$ for any large n. This forces R to be the zero polynomial.

Now, we have $dP(x) = Q(x)B(x)$. It remains to show $d \mid B(x)$ for any $x \in \mathbb{Z}$. Note that $\dfrac{B(a_{d+x})}{d} = \dfrac{P(a_{d+x})}{Q(a_{d+x})}$ is an integer. Also, we have $a_{d+x} = (d+x)! + (d+x) \equiv x \pmod d$. It follows that $B(x) \equiv B(a_{d+x}) \equiv 0 \pmod d$. This finishes the proof. □

Problem 52 (Saudi Arabia IMO TST 2017 Problem 3). Prove that there are infinitely many positive integers n such that n divides $2017^{2017^n-1}-1$ but n does not divide 2017^n-1.

Solution. We claim that all numbers of the form $n = 13 \cdot 2^k$ with $k \in \mathbb{N}$ satisfy the requirements.

Firstly, it is not hard to check that the order of 2017 modulo 13 is 12. As $12 \nmid n$, we have $13 \nmid 2017^n - 1$. Thus, $n \nmid 2017^n - 1$.

Secondly, since $2017 \equiv 1 \pmod{12}$, we have $12 \mid 2017^n - 1$. Therefore, $13 \mid 2017^{2017^n-1} - 1$. Next, by the lifting the exponent lemma for $p = 2$, we have

$$v_2(2017^{2017^n-1} - 1) = v_2(2017^2 - 1^2) + v_2(2017^n - 1) - 1 = 2v_2(2017^2 - 1^2) + v_2(n) - 2 = k + 10.$$

Therefore, $2^k \mid 2017^{2017^n-1} - 1$. Together with $13 \mid 2017^{2017^n-1} - 1$, we obtain $n \mid 2017^{2017^n-1} - 1$ as desired. $\square$